
ECE 592 HW3 - Design of GNNs

— Deadline: 11 Oct 2026 —

Outline

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Task Introduction

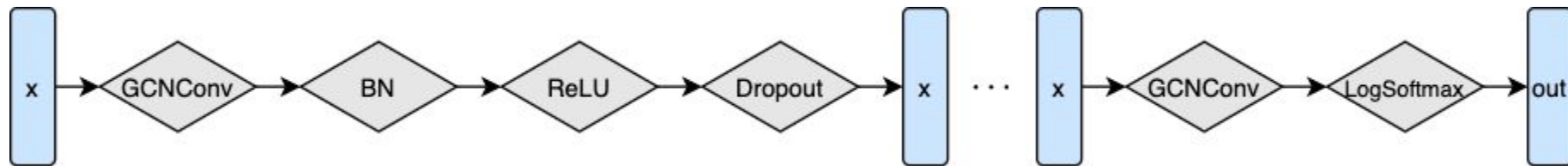
- There are 3 tasks:
 - 1. Inspect PyG dataset and its attributes.
 - 2. Build a GCN from scratch with PyTorch Geometric (PyG).
 - 3. Apply it to two graph learning problems from the Open Graph Benchmark (OGB).
- Run Warmup-Graph.ipynb first. It covers NetworkX and PyG basics.
 - It is not graded so you don't have to submit it.

Task 1: PyG and OGB basics (Q1–Q4)

- Inspect datasets: number of classes, number of features, graph labels, and undirected edge counts.
- Question 1: What is the number of classes and number of features in the ENZYMES dataset? (5 points)
- Question 2: What is the label of the graph with index 100 in the ENZYMES dataset? (5 points)
- Question 3: How many edges does the graph with index 200 have? (5 points)
- Question 4: How many features are in the ogbn-arxiv graph? (5 points)

Task 2: Node property prediction (Q5)

- Predict a property of each node (node classification) in one large graph.
- Implement `GCN.__init__`, `GCN.forward`, `train()`, and `test()`.
- Architecture: `GCNConv` $\rightarrow$ `BatchNorm` $\rightarrow$ `ReLU` $\rightarrow$ `Dropout`, repeated, with `LogSoftmax` at the output.
- Question 5: What are your `best_model` validation and test accuracies? (20 points)



Task 3: Graph property prediction (Q6)

- Predict a property of each whole graph (graph classification).
- Reuse your GCN to produce node embeddings.
- Apply global mean/sum/max pooling, then a linear layer, to get one prediction per graph.
- Question 6: What are your best_model validation and test ROC-AUC scores? (20 points)

Optional (Q7)

1. Try the other global pooling layers in PyG.
2. This is purely for your own understanding, no bonus marks will be given.

Dataset

- Task 1: ENZYMES
 - 600 protein graphs from TUDataset.
 - Used for the data-inspection questions (Q1–Q3).
- Task 2: ogbn-arxiv
 - A citation network of computer science arXiv papers. Each node is a paper; each edge is a citation.
 - Size: 169,343 nodes, 1,166,243 edges, 40 classes.
 - Node features: 128-dimensional, from word embeddings of each paper's title and abstract.
 - Label: the paper's subject area (e.g., cs.LG, cs.AI).
 - Split: by publication year. You train on older papers and test on newer ones.

Dataset (cont.)

- Task 3: ogbg-molhiv
 - 41,127 molecular graphs. Nodes are atoms; edges are chemical bonds.
 - Node features: 9-dimensional atom features, embedded with AtomEncoder.
 - Label: binary, whether the molecule inhibits HIV replication.
 - Split: scaffold split, which groups molecules by structural backbone. Test molecules are structurally different from training molecules.

Evaluation Metrics

- Task 1: Correct outputs shown
- Task 2: Accuracy
 - The fraction of nodes whose predicted class, the argmax of the output, matches the true label.
 - Reported on the train, validation, and test splits.
 - The best model is chosen by validation accuracy, not test accuracy.
- Task 3: ROC-AUC
 - The area under the ROC curve. It measures how well the model ranks positive molecules above negative ones, across all decision thresholds.
 - Used instead of accuracy because the classes are heavily imbalanced. A model that predicts "not active" for every molecule would score high accuracy while being useless.
 - 0.5 means random guessing; 1.0 means perfect ranking.
 - The best model is chosen by validation ROC-AUC.

Evaluation Metrics (cont.)

- Both tasks use the official OGB Evaluator. Don't write your own metric code.
- Reference performance for sanity check:
 - Task 2: $\geq 71\%$ test accuracy.
 - Task 3: $\geq 74\%$ test ROC-AUC.

Grading:

Task 2: 1) Test score $< 60\%$: 0 points, 2) $60-65\%$: 5 points, 3) $65-70\%$: 10 Points, 4) $70-71\%$: 15 points, 5) $71-72\%$: 18 points, 6) $\geq 72\%$: 20 points

Task 3: 1) Test score $< 70\%$: 0 points, 2) $70-74\%$: 5 points, 3) $74-75\%$: 10 points, 4) $75-76\%$: 15 points, 5) $76-77\%$: 18 points, 6) $\geq 77\%$: 20 points

Submission

1. Submit a zip file to Moodle containing:
 - a. Your copy of ECE592_HW3-Design_of_GNNs.ipynb with visible outputs
 - b. A report PDF explaining the results of each question. The report should include:
 - i. Answers to Q1–Q4.
 - ii. Best train, validation, and test scores for Q5 and Q6. (Q7 is optional)
 - iii. Your observations: train/test gaps, why validation and test scores differ etc.
2. Submit one zip file: HW3_<UNITY_ID>.zip
3. Submission Deadline: **11 October 2026, Tuesday**

Grading Policy

- Total: 60 points (15% of final grade)
- The criterion for all questions in each task is that the code should be correctly implemented and runned, AND the corresponding accuracy and/or observations of each question should be provided in the report.
- After completing the specified code changes and reporting the results, you are free to modify the network architecture and record your results (increase/decrease in metric score).
 - This is optional.

Grading Policy (cont.)

Item	Points
Q1: ENZYMES classes and features	5
Q2: Label of graph	5
Q3: Edges in graph	5
Q4: ogbn-arxiv features	5
Q5: Node prediction	20
Q6: Graph prediction	20